Neural Radiance Fields (NeRF) in Home and Interior Design

SEMINAR REPORT

Submitted by

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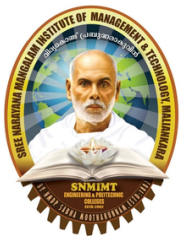
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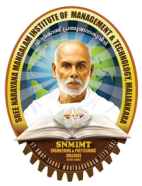
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CERTIFICATE

This is to certify that the seminar report entitled **Neural Radiance Fields (NeRF) in Home and Interior Design** *submitted by* **Sabarinath PS** *(***SNM23CA014***), to the* **Department of Computer Science and Engineering, SNM Institute of Management and Technology, Maliankara**, *in partial fulfilment of the requirement for the award of B.Tech Degree in Computer Science and Engineering (Artificial Intelligence) is a bonafide record of the work carried out by the student.*

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Declaration

I undersigned hereby declare that the seminar report **Neural Radiance Fields (NeRF) in Home and Interior Design**, submitted for partial fulfilment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under the supervision of **Ms. Lakshmi Anand**. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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Abstract

Neural Radiance Fields (NeRF) and advanced 3D spatial representations have revolutionized computer vision, graphics, and indoor spatial computing by enabling the synthesis of photo-realistic 3D environments from sparse 2D imagery. This seminar explores the application of NeRF and related volumetric paradigms—such as 3D Gaussian Splatting—in modern home automation, real estate walkthroughs, and interior design workflows.

We conduct a comprehensive literature review and comparative analysis of four state-of-the-art architectures: 3D-RE-GEN, which facilitates single-image decomposition and modular furniture editing; HouseCrafter, which lifts 2D architectural floorplans into multi-room 3D houses; Gaussian Splatting SLAM, which enables real-time tracking and live dense mapping; and Behind the Veil, which utilizes hierarchical octree structures for occluded surface inpainting.

The discussion highlights how these frameworks overcome legacy CAD and photogrammetry bottlenecks, offering robust solutions for interactive virtual staging, architectural digital twins, and automated spatial design.

Keywords: Neural Radiance Fields; 3D Gaussian Splatting; Indoor Scene Reconstruction; Single-Image 3D Reconstruction; Floorplan-to-3D Generation; Dense SLAM; Occluded Surface Completion; Interior Design; Virtual Staging; Digital Twins.

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List of Abbreviations

3DGS	3D Gaussian Splatting
A-Q	Application-Querying
AR	Augmented Reality
ATE	Absolute Trajectory Error
CAD	Computer-Aided Design
CD	Chamfer Distance
DeCaPE	Depth-enhanced Camera Positional Encoding
DoF	Degrees of Freedom
FID	Fréchet Inception Distance
IoU	Intersection over Union
LPIPS	Learned Perceptual Image Patch Similarity
MLP	Multi-Layer Perceptron
NeRF	Neural Radiance Field
PSNR	Peak Signal-to-Noise Ratio
RANSAC	Random Sample Consensus
RGB-D	Colour and Depth
SDF	Signed Distance Function
SLAM	Simultaneous Localisation and Mapping
SSIM	Structural Similarity Index
TSDF	Truncated Signed Distance Function
VAE	Variational Autoencoder
VFX	Visual Effects
VR	Virtual Reality

Chapter 1

Introduction

Homes and interiors are increasingly designed, sold, and managed through digital 3D models. Buyers take virtual walkthroughs of apartments before visiting them, designers try out furniture arrangements before anything is moved, and smart-home and facility systems rely on *digital twins* of the spaces they control. Building these 3D models, however, has traditionally required manual computer-aided design (CAD) modelling by skilled artists, expensive laser scanners, or photogrammetry pipelines that produce noisy, incomplete meshes [1, 2].

Neural Radiance Fields (NeRF) [5] changed this picture by representing a scene as a continuous function, learned by a neural network, that maps a 3D position and viewing direction to colour and density. From a set of ordinary photographs, NeRF synthesises photorealistic novel views of a scene. Its successor **3D Gaussian Splatting** (3DGS) [6] represents the same kind of radiance field with millions of explicit 3D Gaussians that can be rendered in real time, while diffusion models now generate realistic images and 3D content from little input [7]. Together, these *neural 3D representations* make it possible to capture or generate photorealistic indoor environments from sparse 2D data.

This seminar reviews four recent works that apply these ideas to indoor scenes. Paper [1], the base paper, by Sautter et al., presents **3D-RE-GEN**, which reconstructs a complete, editable 3D room—background and individual furniture—from a single image. Paper [2], by Chen et al., presents **HouseCrafter**, which lifts a 2D floorplan into a multi-room 3D house. Paper [3], by Matsuki et al., presents **Gaussian Splatting SLAM**, which tracks a moving camera and builds a 3DGS map live. Paper [4], by Sun et al., presents **Behind the Veil**, which completes the surfaces hidden behind and beneath furniture.

1.1 General Background

Indoor 3D content can be obtained from different starting points, as Figure 1.1.1 shows. A designer may have only a *single photograph* of a room, an architect may have a *floorplan*, and a surveyor may walk through a building with a *camera* or *depth sensor*. Classical pipelines treat these cases separately and usually output a single, monolithic mesh. The reviewed works instead use learned priors—large pretrained image, diffusion, and 3D models, or geometry learned from many scenes—to turn each kind of input into a usable 3D model: an editable scene with separate objects [1], a complete house [2], a live photorealistic map [3], or a mesh with no holes behind the furniture [4].

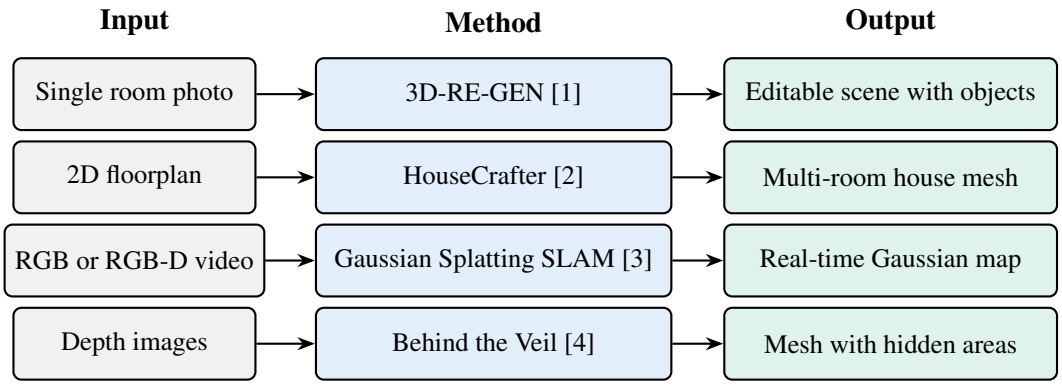


Figure 1.1.1: Inputs and outputs of the reviewed indoor 3D methods

1.2 Motivation

Three needs motivate this study: **casual capture**, since homeowners, agents, and designers want 3D rooms from a phone or a floorplan rather than from expensive scanning [1, 2]; **editability and completeness**, since virtual staging requires moving or swapping furniture without revealing holes or floating objects [1, 4]; and **speed**, since interactive design and AR need models that are built and rendered in real time rather than over hours [3].

1.3 Problem Statement

Given sparse 2D input about an indoor space—a single photograph, a floorplan, a video, or depth images—produce a photorealistic, geometrically complete, and editable 3D model of the space, efficiently enough to support interactive design, real-estate visualisation, and digital-twin applications.

1.4 Objectives

The objectives of this seminar are:

- To introduce 3D scene representations, NeRF, 3D Gaussian Splatting, signed distance fields, diffusion models, and visual SLAM.
- To review in depth four recent works on single-image reconstruction, floorplan lifting, Gaussian Splatting SLAM, and occluded surface completion.
- To compare their inputs, representations, editability, and performance, and to identify open challenges for home and interior design.

1.5 Scope

The seminar covers room- and house-scale indoor scenes, including furniture, walls, and floors, reconstructed or generated from images, floorplans, video, and depth, together with the datasets and metrics used to evaluate them. Outdoor scenes and dynamic humans are outside its scope.

1.6 Advantages

Neural 3D representations offer several advantages over legacy CAD and photogrammetry workflows:

- **Photorealism:** radiance fields capture reflections, soft lighting, and fine detail.
- **Low Capture Cost:** a phone camera, a single image, or a floorplan can be enough.
- **Completeness:** learned priors fill in regions that were never observed.
- **Interactivity:** Gaussian Splatting renders scenes in real time for walkthroughs and AR.

1.7 Organization of the Report

Chapter 2 introduces the theoretical foundations. **Chapter 3** reviews the four works in detail. **Chapter 4** compares the approaches, **Chapter 5** discusses applications, **Chapter 6** presents challenges and future directions, and **Chapter 7** concludes the report.

Chapter 2

Theoretical Foundations

This chapter introduces the concepts on which the reviewed works build: 3D scene representations, neural radiance fields, 3D Gaussian Splatting, signed distance fields and mesh extraction, diffusion models for 3D content, visual SLAM, and the evaluation metrics used in indoor reconstruction.

2.1 3D Scene Representations

A 3D scene can be stored **explicitly**, as point clouds, voxel grids, or triangle meshes, or **implicitly**, as a function that a neural network evaluates at any 3D point [4]. Meshes are compact and are what CAD tools, game engines, and physics simulators expect, but fusing new observations into them is difficult. Voxel grids are easy to update but use much memory at fixed resolution. Implicit fields are continuous and can fill small holes, but must be queried point by point. Gaussian splats combine an explicit, flexible set of primitives with smooth, differentiable rendering [3]. Table 2.1 compares these options.

Table 2.1: 3D scene representations for indoor reconstruction

Representation	**Strength**	**Weakness**	**Used in**
Triangle mesh	Editable; standard for CAD, games, physics	Hard to fuse and optimise	[1, 2, 4]
Voxel grid / TSDF	Simple fusion of depth	Memory-hungry; fixed resolution	[2]
Neural implicit field	Continuous; fills small holes	Slow per-point queries	[4]
3D Gaussians	Real-time, differentiable rendering	No explicit surface	[3]

2.2 Neural Radiance Fields

A **Neural Radiance Field** [5] represents a scene with a multi-layer perceptron (MLP) F_Θ that maps a 3D position $\mathbf{x} = (x, y, z)$ and viewing direction (θ, ϕ) to a colour $\mathbf{c}$ and a volume density σ. A pixel is rendered by marching along its camera ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$ and accumulating colour weighted by density:

$$\hat{C}(\mathbf{r}) = \sum_{i=1}^{N} T_i (1 - e^{-\sigma_i \delta_i}) \mathbf{c}_i, \qquad T_i = \exp\Big(- \sum_{j<i} \sigma_j \delta_j \Big), \tag{2.1}$$

where δ_i is the distance between samples and T_i the transmittance. Because this rendering is differentiable, F_Θ is trained simply by minimising the difference between rendered and observed pixels across many photographs. NeRF captures view-dependent effects and fine detail, but its per-pixel ray marching makes training and rendering slow [3].

2.3 3D Gaussian Splatting

3D Gaussian Splatting [6] replaces the MLP with a large set of anisotropic 3D Gaussians, each with a mean $\boldsymbol{\mu}$, covariance Σ, colour c_i, and opacity α_i. To render a view with camera pose T_{CW}, each Gaussian is projected to a 2D Gaussian on the image plane,

$$\boldsymbol{\mu}_I = \pi(T_{CW}\, \boldsymbol{\mu}_W), \qquad \Sigma_I = J W \Sigma_W W^\top J^\top, \tag{2.2}$$

where π is the projection, W the rotation of T_{CW}, and J the Jacobian of the projection, and the sorted Gaussians are alpha-blended:

$$C_p = \sum_{i \in \mathcal{N}} c_i\, \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j). \tag{2.3}$$

Because it *rasterises* primitives instead of marching rays, 3DGS skips empty space and renders at hundreds of frames per second while remaining fully differentiable [3].

2.4 Signed Distance Fields and Mesh Extraction

A **signed distance function** (SDF) gives, for every point, its distance to the nearest surface, negative inside objects and positive outside; the surface is its zero level set. Neural SDFs such

as DeepSDF [8] learn shapes as continuous functions. In **TSDF fusion** [9], depth images are integrated into a voxel grid of truncated signed distances by weighted averaging, as in KinectFusion [10]. The **Marching Cubes** algorithm [11] then extracts a triangle mesh from the zero crossings of any SDF grid. HouseCrafter fuses its generated RGB-D images with TSDF fusion [2], and Behind the Veil predicts SDFs with neural decoders before running Marching Cubes [4].

2.5 Diffusion Models for 3D Content

Diffusion models learn to generate images by reversing a gradual noising process. **Latent diffusion**, as in Stable Diffusion [7], performs this denoising in the compressed latent space of a variational autoencoder (VAE), using a UNet with attention layers that can be conditioned on text, images, or layouts. Because 3D data is scarce while 2D images are abundant, many 3D methods fine-tune such 2D models to produce consistent *multi-view* images, which are then fused into 3D [2], or use image-editing and image-to-3D models as components of a larger pipeline [1].

2.6 Visual SLAM

Simultaneous Localisation and Mapping (SLAM) estimates a camera's trajectory while building a map of its surroundings from a video stream. The camera pose at each frame is a rigid transformation $T_{CW} \in SE(3)$, optimised in its Lie algebra so that updates stay valid rotations. *Tracking* estimates the pose of each new frame against the map, and *mapping* refines the map from a window of selected *keyframes*. Monocular SLAM uses a single colour camera and must also infer depth; RGB-D SLAM uses a depth sensor. Map-centric systems maintain one global 3D representation, such as voxels, surfels, or neural fields, and track against it [3].

2.7 Evaluation Metrics

The reviewed works evaluate geometry, appearance, and tracking with different metrics (Table 2.2). **Accuracy** and **completeness** measure how much of the predicted surface lies near the ground truth and vice versa, and their harmonic mean is the **F-score**; **Chamfer** and **Hausdorff** distances measure average and worst-case point distances [1, 4]. Rendered images are compared with PSNR, SSIM, and LPIPS [3], and generated images are judged with FID and

Inception Score [2]. Camera tracking is measured by the absolute trajectory error (ATE) [3].

Table 2.2: Evaluation metrics used in the reviewed works

Metric	Measures	Used in
Chamfer, Hausdorff, F-score, IoU	Geometric agreement of scenes and object boxes	[1]
Accuracy, completeness, F1	Surface quality, including hidden regions	[4]
PSNR, SSIM, LPIPS	Quality of rendered views	[1, 2, 3]
FID, Inception Score	Realism of generated images	[2]
ATE RMSE	Camera tracking error	[3]
User studies	Perceived quality and layout	[1, 2]

2.8 Taxonomy of Approaches

Figure 2.8.1 organises the reviewed works by the task they solve. Two *generate* 3D scenes from limited 2D input—one image [1] or a floorplan [2]—and two *reconstruct* scenes from measurements—a live video [3] or depth images whose hidden regions must be completed [4].

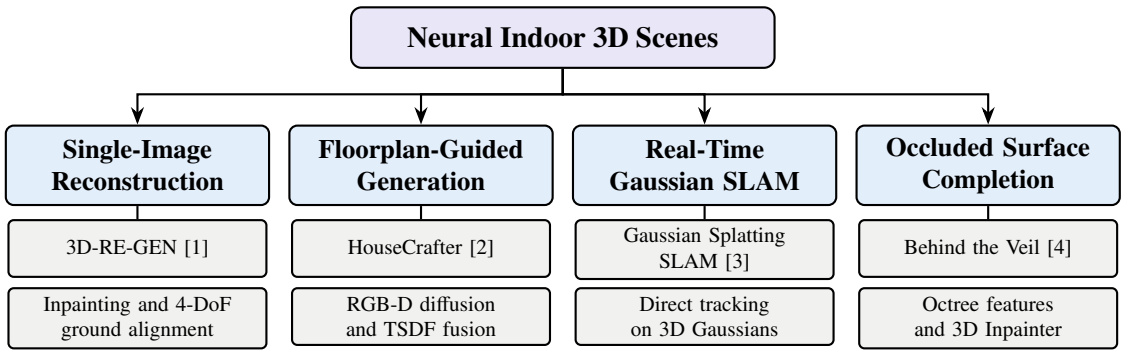


Figure 2.8.1: Taxonomy of the reviewed approaches

Chapter 3

Literature Survey

This chapter reviews the four works in detail. For each, it describes the problem and motivation, the method, the experimental setup and results, and closes with a short discussion of strengths and limitations.

3.1 3D-RE-GEN: 3D Reconstruction of Indoor Scenes with a Generative Framework

The base paper, by Sautter, Dihlmann, and Lensch [1], from the University of Tübingen, presents **3D-RE-GEN**, a compositional framework that turns a single photograph of a room into a textured background mesh and a set of separate, textured 3D objects that can be moved, relit, and simulated (Figure 3.1.1).



Figure 3.1.1: 3D-RE-GEN decomposes a single image into an editable scene [1]

3.1.1 Problem Statement and Motivation

Building a 3D interior by hand is a production bottleneck: the room, every piece of furniture, and the final layout are each modelled by specialised artists. Single-image-to-3D models

such as SF3D and Hunyuan3D work well for isolated objects but fail on cluttered multi-object scenes, where the image lacks depth, objects hide one another, and nothing states their distances, scale, or contact points [1]. Earlier holistic, retrieval-based, and compositional methods share two gaps: the *background* of walls and floor is usually ignored, and nothing enforces *physical plausibility*, so objects float above or sink into the floor.

3.1.2 Method

3D-RE-GEN chains strong pretrained models and adds two novel contributions, namely **Application-Querying** for occluded objects and a **4-DoF ground-constrained optimisation** for layout (Figure 3.1.2).

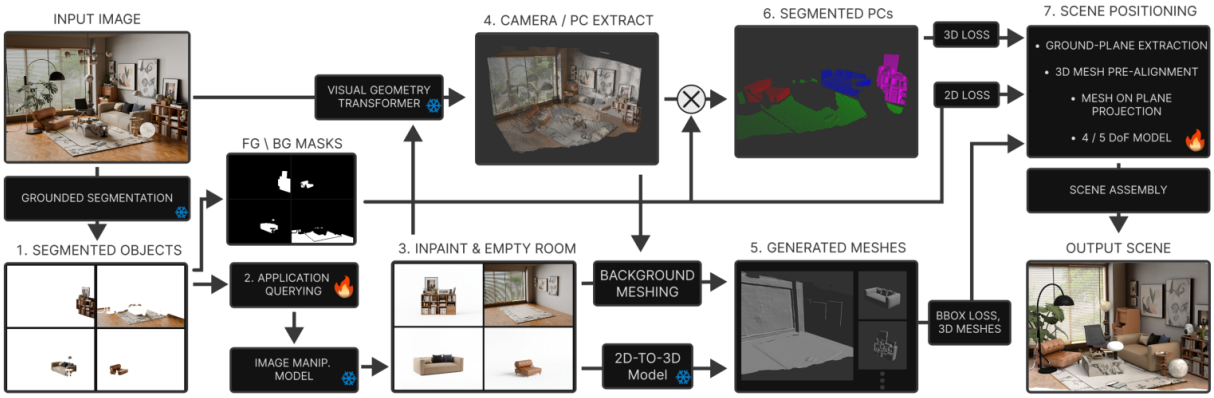
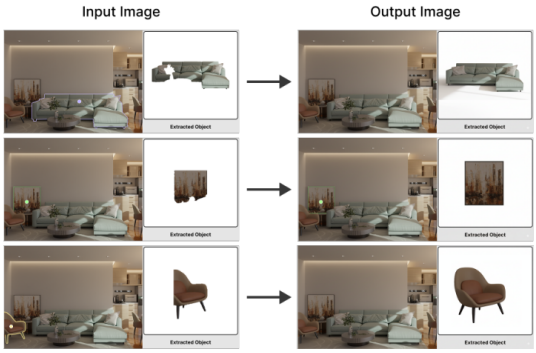


Figure 3.1.2: Overview of the 3D-RE-GEN framework [1]

Segmentation. Grounded SAM performs text-prompted instance segmentation of the input image I, and a lightweight tool lets the user correct the masks M_{obj}, which later serve as 2D silhouette targets [1].

Application-Querying (A-Q). Asking a generative model to complete a partially hidden object often fails because it lacks scene context. A-Q builds a composite query image styled like a user interface with two panels: the full image with the target outlined, and the occluded segment on a white background. An image-editing model (Gemini 2.5 Flash Image) is asked to return the same interface with the object completed, so it infers hidden parts with consistent perspective, lighting, and material (Figure 3.1.3). The same model removes all objects to produce an "empty room" image I_{bg} [1].



(a) Input and output of the Application-Querying (A-Q) method.



Figure 3.1.3: Application-Querying with a GUI-style visual prompt [1]

Scene, camera, and assets. The geometry transformer VGGT receives both I and I_{bg} and returns aligned cameras and point clouds; the background cloud is meshed into a background usable for rendering and simulation, and back-projected masks cut per-object target clouds P_{target} from the scene cloud. Each completed object image is converted into a textured mesh by Hunyuan3D 2.0 [1].

Differentiable placement. Objects whose masks touch the floor mask use the **4-DoF planar model**; others use a 5-DoF model (translation, yaw, scale). The floor plane is fitted with RANSAC, and in its local frame the planar model learns only a 2D translation (t'_x, t'_z), a yaw r'_y, and a scale s:

$$V_{world} = T_{plane \to world}\big(s\, R_y(r'_y)\, V_{local} + (t'_x,\, 0,\, t'_z)^\top\big), \tag{3.1}$$

so objects can only slide and rotate on the floor. Both models minimise

$$\mathcal{L}_{total} = w_{sil}\, \mathcal{L}_{silhouette} + w_{3D}\, \mathcal{L}_{3D} + w_{bbox}\, \mathcal{L}_{bbox}, \tag{3.2}$$

combining a Dice and Focal silhouette loss, a point-to-face distance to P_{target}, and a penalty on vertices that leave the background bounding box [1].

3.1.3 Experimental Setup

The training-free pipeline uses PyTorch3D for differentiable rendering; a room with about ten objects takes 17–20 minutes on one GPU, or 7–8 minutes on four. Evaluation uses hand-picked synthetic CGTrader scenes, unseen by competing methods trained on 3D-FRONT, plus real photographs. Scenes are normalised and aligned with ICP, then compared with Chamfer distance, F-score, bounding-box IoU, and Hausdorff distance against MIDI and DepR [1].

3.1.4 Results and Analysis

As Table 3.1 shows, 3D-RE-GEN outperforms both baselines on every metric: its Chamfer distance of 0.011 is less than a third of MIDI's, its F-score reaches 0.85, its IoU of 0.63 shows correct relative object placement, and its Hausdorff distance of 0.33 indicates few outliers or floating artefacts. MIDI often merges or duplicates objects and DepR produces misaligned "blobs". In a study with 59 participants, 3D-RE-GEN received 81% of the preferences, mostly for its layout [1].

Table 3.1: 3D-RE-GEN against MIDI and DepR

Method	**CD ↓**	**F-Score ↑**	**IoU ↑**	**Precision ↑**	**Recall ↑**	**Hausdorff ↓**
DepR	0.028	0.65	0.44	0.12	0.18	0.61
MIDI	0.036	0.70	0.57	0.15	0.13	0.55
3D-RE-GEN	**0.011**	**0.85**	**0.63**	**0.21**	**0.19**	**0.33**

Ablation. Without the 4-DoF model, objects drift off the floor and Chamfer distance rises to 0.019; without A-Q, the 3D generator sees only occluded segments and the F-score drops to 0.68, with sharply worse perceptual scores (Table 3.2) [1].

Table 3.2: Ablation of the 4-DoF model and Application-Querying

Variant	**CD ↓**	**F-Score ↑**	**IoU ↑**	**Hausdorff ↓**	**SSIM ↑**	**LPIPS ↓**
Without 4-DoF	0.019	0.82	0.52	0.51	0.27	0.46
Without A-Q	0.030	0.68	0.51	0.52	0.13	0.66
Full 3D-RE-GEN	**0.011**	**0.85**	**0.63**	**0.33**	**0.29**	**0.43**

3.1.5 Discussion

3D-RE-GEN outputs exactly what designers need—separate textured furniture on a complete background—and is training-free and modular. Its limitations are its dependence on several

large, partly proprietary models, minutes of runtime per room, occasional manual mask correction, a small evaluation set, and a ground-plane assumption that does not cover wall-mounted or stacked objects.

3.2 HouseCrafter: Lifting Floorplans to 3D Scenes with 2D Diffusion Models

Chen, Nguyen, Voleti, Jampani, and Jiang [2], from Northeastern University and Stability AI, present **HouseCrafter**, which turns a 2D floorplan—walls, doors, windows, and furniture boxes—into a complete, textured 3D house mesh using a pretrained image diffusion model (Figure 3.2.1).

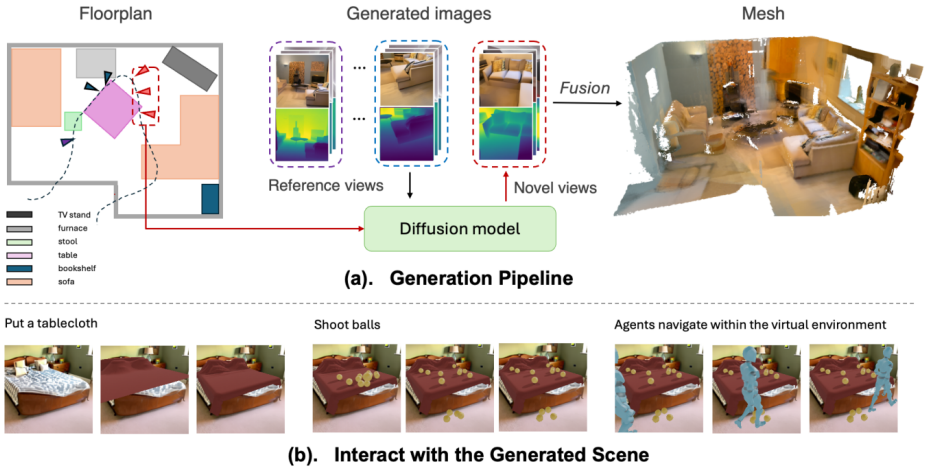


Figure 3.2.1: HouseCrafter generation pipeline and scene interaction [2]

3.2.1 Problem Statement and Motivation

3D data is scarce compared with web-scale images, so training 3D generators directly is hard. Multi-view diffusion models work for single objects, whose views fit in one batch, but not for houses. Text-guided methods such as Text2Room build rooms image by image with monocular depth, suffering scale ambiguity and lacking global structure, and 3D proxy boxes become cumbersome for large scenes. A 2D floorplan is a familiar, intuitive way to specify a whole house [2].

3.2.2 Method

Autoregressive generation. Camera locations sampled over the free space of the floorplan form a graph that is traversed batch by batch: images already generated at visited nodes within r hops serve as references for novel views at unvisited nodes within n hops, and the first batch is conditioned on the floorplan alone. TSDF fusion finally integrates all RGB-D images into a mesh [2].

Joint RGB-D diffusion. The UNet of Stable Diffusion v1.5 [7] is fine-tuned with its VAE frozen (Figure 3.2.2). RGB and depth latents are concatenated, $z = [x, d]$, and denoised jointly; depth is clipped to 0–3 metres and replicated to three channels for the VAE. Generating *metric* depth avoids the scale ambiguity of monocular estimators [2].

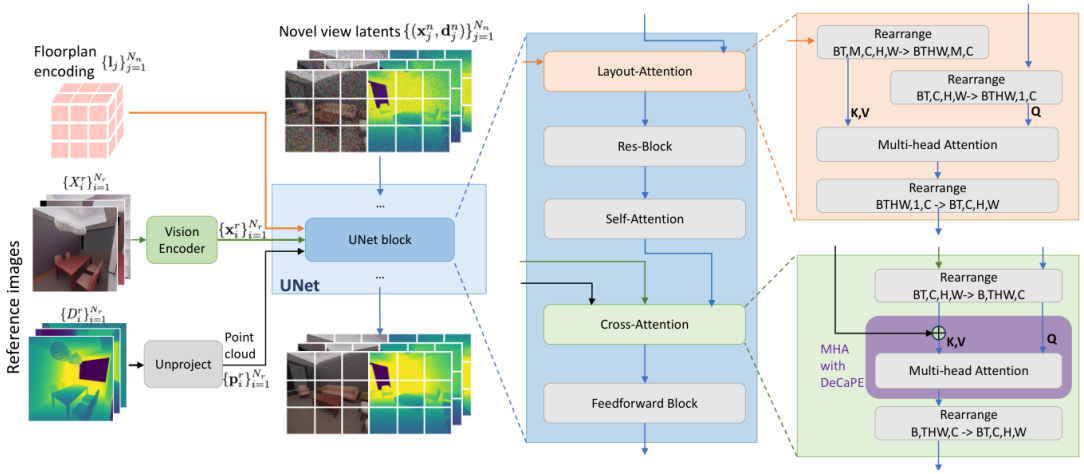


Figure 3.2.2: Floorplan-guided RGB-D generation model [2]

Floorplan conditioning. For every latent pixel, the camera ray is projected onto the floor plane and up to M intersections with floorplan components are collected, stopping at the first wall. Each intersection's category c and position p are embedded as $l = \text{Embed}(c) + \text{PosEnc}(p)$, and a **layout-attention** layer at the start of each UNet block lets every pixel attend to the floorplan along its own ray [2].

Depth-enhanced camera encoding. Novel views attend to reference views through cross-attention. Extending camera positional encoding (CaPE), **DeCaPE** adds the 3D position p_K of

each reference token, obtained from its depth:

$$s_{QK} = v_Q^\top \phi(\pi_Q^{-1}\pi_K)(v_K + \text{PosEnc}(p_K)), \tag{3.3}$$

which depends only on relative poses and makes attention geometry-aware [2].

3.2.3 Experimental Setup

About 2,000 3D-FRONT houses with 2 million renders are used for training and 300 for evaluation. Metrics are FID and Inception Score; PSNR and depth error between reference and novel (R-N) and between novel (N-N) views; floorplan consistency, measured by segmenting the scene with ODIN and computing mAP@25 against the floorplan boxes; and the number of dangling components. Baselines are CC3D, BlockFusion with Meshy textures, and Text2Room, and a 12-person user study is added [2].

3.2.4 Results and Analysis

HouseCrafter achieves the highest floorplan consistency (Table 3.3), with mAP@25 of 45.77 against 25.60 for CC3D, approaching 54.51 for ground-truth images. Text2Room and BlockFusion score higher Inception Scores, which the authors attribute to models trained on real images; even ground-truth renders score only 4.50. Users strongly preferred HouseCrafter for quality and floorplan consistency [2].

Table 3.3: HouseCrafter against scene generation baselines

Method	**IS ↑**	**mAP@25 ↑**	**Artifacts ↓**
Text2Room	**5.35**	–	302.5
CC3D	4.02	25.60	–
BlockFusion + Meshy	5.01	0.81	**118.64**
HouseCrafter	4.37	**45.77**	178.0
Ground-truth images	4.50	54.51	–

Ablation. Generating depth with RGB lowers FID from 49.35 to 33.39; floorplan conditioning is the largest gain, cutting FID to about 16 and N-N depth error below 8; and DeCaPE gives the lowest depth errors and raises mAP@25 to 52.26 (Table 3.4) [2].

Table 3.4: Ablation of HouseCrafter design choices

Variant	FID ↓	PSNR ↑ R-N	AbsRel ↓ R-N	AbsRel ↓ N-N	mAP@25 ↑
RGB only	49.35	–	–	–	–
+ Output depth	33.39	20.99	23.56	11.48	–
+ Input depth	35.77	20.91	22.28	12.05	–
+ Floorplan (no input depth)	**15.64**	**25.36**	7.65	7.85	48.46
+ Floorplan and input depth	16.70	24.47	7.73	8.20	49.74
+ DeCaPE (full)	17.75	25.31	**6.79**	**7.37**	**52.26**

3.2.5 Discussion

HouseCrafter's strength is its interface: a floorplan, which architects and agents already have, controls a whole house, and the output is an explicit mesh. Reusing an image diffusion model avoids large 3D training sets. Its limitations are training on synthetic houses, a coarse vertex-coloured mesh not separated into objects, many diffusion steps per house, and floorplans that specify footprints but not style or height.

3.3 Gaussian Splatting SLAM

Matsuki, Murai, Kelly, and Davison [3], from Imperial College London, present the first SLAM system that uses **3D Gaussian Splatting** as its only scene representation. From a single moving camera it tracks the camera pose and builds a photorealistic Gaussian map live at about 3 frames per second, capturing thin wires and even transparent glass, and it extends naturally to RGB-D input.

3.3.1 Problem Statement and Motivation

Every dense SLAM map has a weakness: voxel grids are memory-hungry with bounded resolution, meshes need irregular topology updates, surfels are hard to optimise, and neural fields need expensive per-pixel ray casting. 3DGS [6] is as flexible as a point cloud yet continuous and differentiable, rendering up to 200 frames per second at 1080p, but it had been used only offline, with camera poses computed beforehand by structure from motion. A live system must estimate poses and grow the map simultaneously [3].

3.3.2 Method

Tracking, keyframing, and mapping share one Gaussian map (Figure 3.3.1). Colour is alpha-blended as in Chapter 2, and depth likewise, $D_p = \sum_i z_i \alpha_i \prod_{j<i}(1-\alpha_j)$ [3].

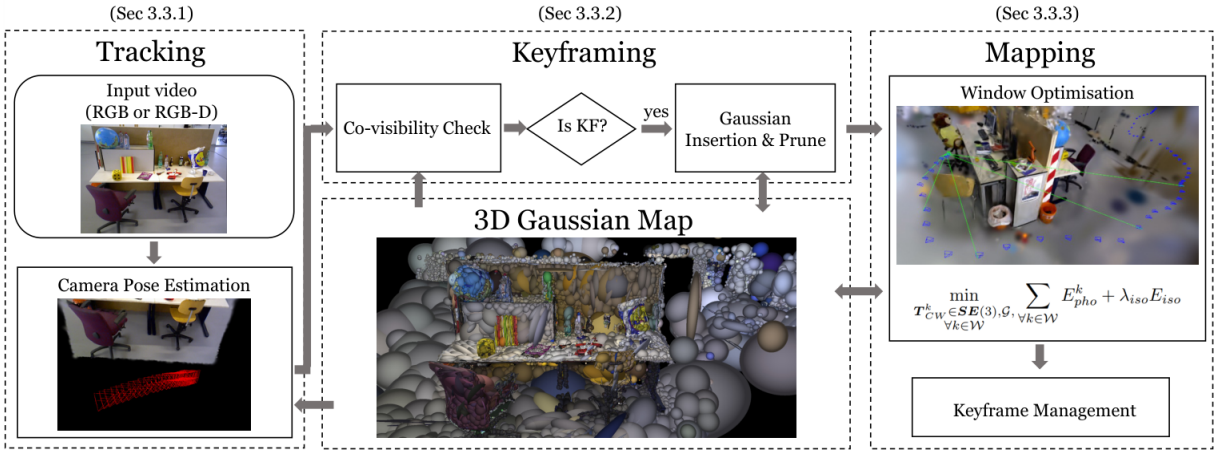


Figure 3.3.1: Overview of the Gaussian Splatting SLAM system [3]

Tracking. For every new frame only the camera pose T_{CW} is optimised, with the map fixed, by minimising the photometric residual

$$E_{pho} = \left\| I(\mathcal{G}, T_{CW}) - \bar{I} \right\|_1, \tag{3.4}$$

plus a depth residual $E_{geo} = \|D(\mathcal{G}, T_{CW}) - \bar{D}\|_1$ when depth is available. As tracking needs at least 50 iterations per frame, the authors derive the first analytic Jacobians of projected Gaussians with respect to the pose on the Lie group $SE(3)$, for example

$$\frac{D\boldsymbol{\mu}_C}{DT_{CW}} = \begin{bmatrix} \mathbf{I} & -\boldsymbol{\mu}_C^{\times} \end{bmatrix}, \tag{3.5}$$

where $^{\times}$ is the skew-symmetric matrix, implemented in CUDA inside the rasteriser [3].

Keyframing. A small keyframe window $\mathcal{W}_k$ is optimised. A Gaussian is visible in a view if it is rasterised before the accumulated opacity reaches 0.5, so occlusion is handled naturally; a frame becomes a keyframe when the IoU of its visible Gaussians with the last keyframe drops below a threshold. New Gaussians are inserted at keyframes from measured or rendered depth,

and recent Gaussians not seen by at least three other frames are pruned [3].

Mapping. Mapping refines the Gaussians and keyframe poses in the window plus two random past keyframes. Because rasterisation does not constrain a Gaussian's extent along the viewing ray, Gaussians tend to become long needles that create artefacts and disturb tracking, so an **isotropic regularisation** penalises each Gaussian's scales $\mathbf{s}_i$ for differing from their mean $\tilde{s}_i$:

$$E_{iso} = \sum_{i=1}^{|\mathcal{G}|} \left\| \mathbf{s}_i - \tilde{s}_i \mathbf{1} \right\|_1, \qquad \min_{T_{CW}^k, \mathcal{G}} \sum_{k \in \mathcal{W}} E_{pho}^k + \lambda_{iso} E_{iso}, \tag{3.6}$$

with geometric residuals added when depth is available [3].

3.3.3 Experimental Setup

Evaluation uses three real TUM RGB-D sequences (monocular and RGB-D), eight synthetic Replica sequences (RGB-D), and live RealSense D455 captures, on an Intel i9-12900K with an RTX 4090. Tracking is measured by ATE RMSE and maps by PSNR, SSIM, and LPIPS on non-keyframes, averaged over three runs. Baselines without loop closure are DSO, DROID-VO, and DepthCov-VO for monocular input, and neural-field SLAM systems such as NICE-SLAM, ESLAM, Co-SLAM, and Point-SLAM for RGB-D input [3].

3.3.4 Results and Analysis

Tracking. On TUM (Table 3.5), with a single camera and no learned depth prior, the method averages 3.96 cm against 7.73 cm for DROID-VO; with RGB-D input it averages 1.47 cm, the best among compared methods and close to ORB-SLAM2 with loop closure (Figure 3.3.2). On Replica, its single-process version averages 0.32 cm [3].

Table 3.5: Camera tracking error on the TUM RGB-D dataset

Input	Method	fr1/desk	fr2/xyz	fr3/office	Avg. (cm)
Monocular	DSO	22.4	**1.10**	9.50	11.0
	DROID-VO	5.20	10.7	7.30	7.73
	DepthCov-VO	5.60	1.20	68.8	25.2
	Gaussian Splatting SLAM	**3.78**	4.60	**3.50**	**3.96**
RGB-D	NICE-SLAM	4.26	6.19	3.87	4.77
	ESLAM	2.47	**1.11**	2.42	2.00
	Point-SLAM	4.34	1.31	3.48	3.04
	Gaussian Splatting SLAM	**1.50**	1.44	**1.49**	**1.47**

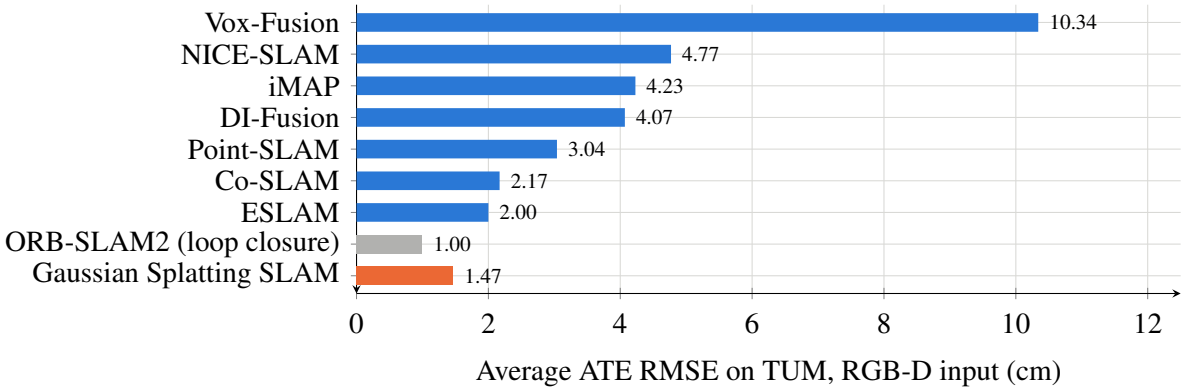


Figure 3.3.2: Average tracking error of RGB-D SLAM methods (data from [3])

Rendering and memory. On Replica (Table 3.6) the method reaches 38.94 dB PSNR and the lowest LPIPS while rendering at 769 frames per second, hundreds of times faster than neural-field systems; Point-SLAM's slightly higher SSIM relies on ground-truth depth. The map needs only 2.6 MB for monocular and 3.97 MB for RGB-D input on TUM, against 6.4 MB for Co-SLAM [3].

Table 3.6: Novel view rendering on the Replica dataset

Method	**PSNR (dB) ↑**	**SSIM ↑**	**LPIPS ↓**	**Rendering FPS ↑**
NICE-SLAM	24.42	0.809	0.233	0.54
Vox-Fusion	24.41	0.801	0.236	2.17
Point-SLAM	35.17	**0.975**	0.124	1.33
Gaussian Splatting SLAM	**38.94**	0.968	**0.070**	**769**

Ablation and convergence. Removing isotropic regularisation raises monocular ATE from 3.96 to 4.83 cm, and removing keyframe selection raises it to 8.73 cm. Cameras placed up to 1.2 m from a target converge on a Gaussian map in about 80% of trials, against 33% for an MLP SDF and 14% for a hash-grid SDF [3].

3.3.5 Discussion

Gaussian Splatting SLAM shows that a photorealistic model of a room can be built *while* walking through it with an ordinary camera and viewed in real time, with one compact representation for tracking, mapping, and rendering. Its limitations are the lack of loop closure for large homes, about 3 frames per second on a high-end GPU in monocular mode, no clean surface mesh or object separation, and empty regions wherever the camera did not look.

3.4 Behind the Veil: Enhanced Indoor 3D Scene Reconstruction with Occluded Surfaces Completion

Sun, Zhao, Guo, Wang, Huang, Chen, and Ren [4], from Purdue University and Bosch Research North America, present a reconstruction method that, from a sequence of depth images, recovers not only the visible surfaces of a room but also the surfaces hidden by occlusion—the back of a sofa, the floor under a bed, and the wall behind a television.

3.4.1 Problem Statement and Motivation

Virtual touring and room rearrangement need re-configurable scenes: when a sofa is moved, its back and the floor beneath it must not appear as holes. TSDF fusion [9] and surfels are limited by depth noise and blocked views, neural implicit methods such as BNV-Fusion fill only small gaps, and object-centric priors need category-specific pretraining that cannot cover every object in a room [4].

3.4.2 Method

Coarse-fine octree. The unprojected depth points are stored in an L-level octree with learnable features at node corners, interpolated trilinearly (Figure 3.4.1). With $j = 4$ and $L = 9$, the levels split into *coarse* features $O_c = \{O_i\}_{i=0}^{j}$, encoding contextual structure that generalises to unseen regions, and *fine* features $O_f = \{O_i\}_{i=j+1}^{L}$, encoding detailed visible geometry [4].

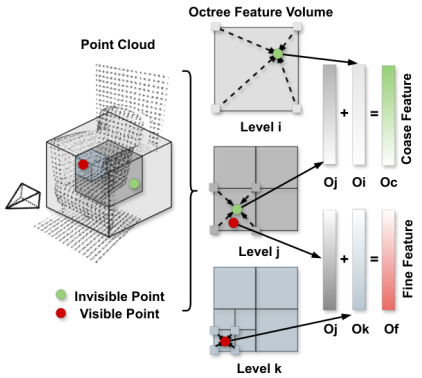


Figure 3.4.1: Coarse-fine hierarchical octree feature volume [4]

Dual decoders. Two decoders predict signed distances: a scene-specific **Geo-decoder** D_{Geo}, fed with fine features, for visible surfaces, and a cross-scene **3D Inpainter** D_{Ip}, fed with coarse

features, for occluded surfaces. With f a positional encoding,

$$d_{p_v} = D_{Geo}(f(p_v), O_f(p_v)), \qquad d_{p_i} = D_{Ip}(f(p_i), O_c(p_i)). \tag{3.7}$$

The method runs in three stages (Figure 3.4.2) [4].

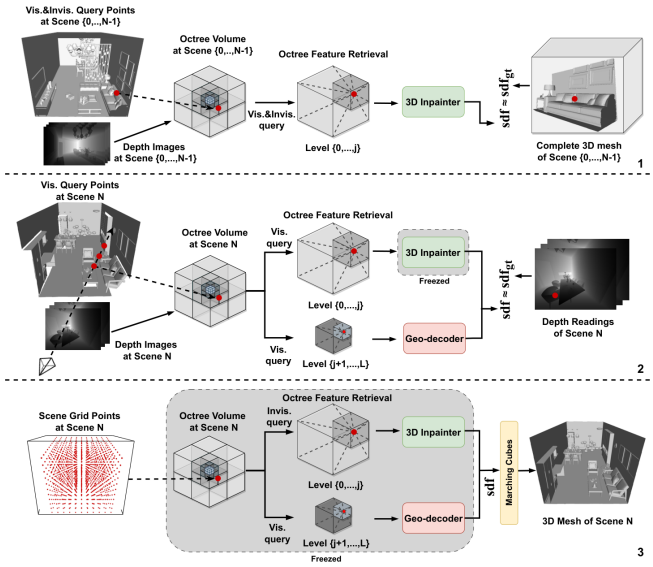


Figure 3.4.2: Three-stage pipeline of Behind the Veil [4]

Stage 1: training the 3D Inpainter. Points with ground-truth signed distances are sampled from complete training meshes. In a decoder-only latent-optimisation design, each scene keeps its own octree features while all share the Inpainter, trained with a binary cross-entropy on sigmoid-mapped SDFs plus eikonal and smoothness terms [4]:

$$\mathcal{L} = \mathcal{L}_{bce} + \lambda_{eik}(1 - \|\nabla D_{Ip}\|)^2 + \lambda_{smooth}\left\|\nabla D_{Ip}(p) - \nabla D_{Ip}(p + \epsilon)\right\|^2. \tag{3.8}$$

Stage 2: fitting a new scene. With only depth images available, points sampled along camera rays near observed surfaces supervise the Geo-decoder and both feature sets, while the Inpainter stays frozen, so the coarse features adapt until the Inpainter explains the visible surfaces and continues them into hidden regions [4].

Stage 3: mesh extraction. Grid points lacking features at more than $\alpha = 3$ octree levels are treated as invisible and sent to the Inpainter, the rest to the Geo-decoder, and Marching Cubes

[11] extracts the complete mesh [4].

3.4.3 Experimental Setup

The authors build the **3D Completed Room Scene** (3D-CRS) dataset of 20 rooms with complete meshes (15 for training, 5 for testing) and also use iTHOR scenes (6 test scenes) and ScanNet for qualitative tests. Accuracy, completeness, and F1 are compared with TSDF Fusion, Go-Surf, and BNV-Fusion [4].

3.4.4 Results and Analysis

Completeness rises from 76.5 for BNV-Fusion to 89.3 on 3D-CRS and from 72.3 to 89.8 on iTHOR, giving the best F1 on every test scene (Table 3.7). Accuracy is slightly below BNV-Fusion's because it also covers the harder inpainted regions. Baselines leave holes behind sofas and under furniture, whereas Behind the Veil completes planar areas and thin structures alike, leaving a complete room layout even after furniture is removed [4].

Table 3.7: Average reconstruction quality on 3D-CRS and iTHOR

Method	**3D-CRS**			**iTHOR**		
	Accu.	**Comp.**	**F1**	**Accu.**	**Comp.**	**F1**
TSDF Fusion	84.6	60.2	70.2	81.3	66.1	72.9
Go-Surf	90.1	70.8	79.3	88.9	69.8	78.2
BNV-Fusion	**92.5**	76.5	83.7	**91.3**	72.3	80.7
Behind the Veil	90.4	**89.3**	**89.9**	88.8	**89.8**	**89.3**

Fewer depth frames. With only 50 depth frames per scene, BNV-Fusion's completeness falls from 76.5 to 62.3, whereas Behind the Veil stays above 87.5 because its Inpainter fills what the frames miss (Figure 3.4.3) [4].

Ablation. Without a cross-scene Inpainter, completeness stays near BNV-Fusion's level; coarse features alone lose accuracy and fine features alone lose completeness (Table 3.8) [4].

3.4.5 Discussion

Behind the Veil solves a problem central to virtual staging: once furniture moves, the surfaces behind and beneath it must exist. Its split between a scene-specific decoder and a shared Inpainter, trained on only 15 rooms, generalises even to real ScanNet data. Its limitations are

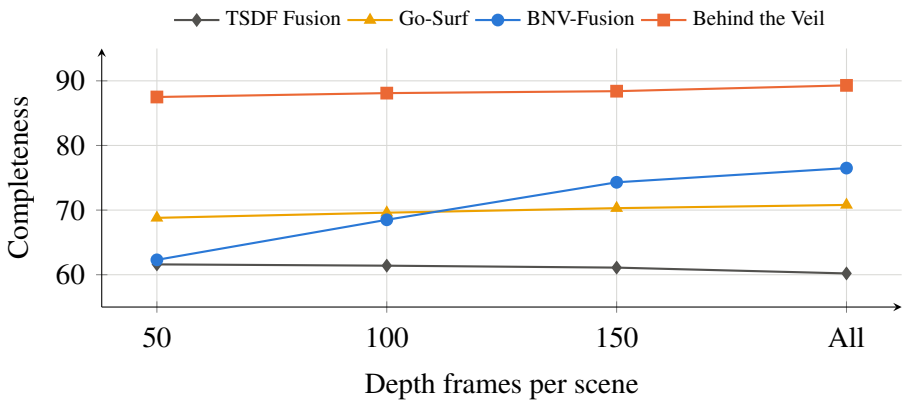


Figure 3.4.3: Completeness versus number of depth frames (data from [4])

Table 3.8: Ablation of the 3D Inpainter and coarse-fine features

Variant	Accu.	Comp.	F1
Without 3D Inpainter	**91.4**	82.2	86.5
Without cross-scene training	83.3	82.4	82.9
Only coarse features	86.0	88.5	87.2
Only fine features	86.6	82.2	84.3
Full method	89.1	**90.1**	**89.7**

geometry without colour, the need for posed depth images, completed surfaces that are plausible guesses, and per-scene optimisation that is not real-time.

3.5 Summary of the Literature Survey

Table 3.9 summarises the four reviewed works.

Table 3.9: Summary of the reviewed works

Work	Venue	Input	Key idea	Output
3D-RE-GEN [1]	arXiv 2025	One RGB image	Application-Querying inpainting and 4-DoF ground-aligned placement of generated assets	Textured objects on a background mesh
HouseCrafter [2]	ICCV 2025	2D floorplan	Floorplan-guided multi-view RGB-D diffusion with DeCaPE, fused by TSDF	Multi-room house mesh
Gaussian Splatting SLAM [3]	CVPR 2024	RGB or RGB-D video	Analytic $SE(3)$ Jacobians, isotropic regularisation, and Gaussian keyframing	Live 3D Gaussian map and camera path
Behind the Veil [4]	CVPR 2025	Depth images with poses	Coarse-fine octree with Geo-decoder and cross-scene 3D Inpainter	Complete mesh including hidden surfaces

Chapter 4

Discussion and Comparison

This chapter compares the four works along the dimensions that matter for home and interior design: what input they need, how they represent the scene, how editable and complete the result is, how fast they run, and how they are evaluated.

4.1 Comparison of Methodologies

Table 4.1 compares the methods. Two are *generative*—they invent content the input does not show, using large pretrained models—and two are *reconstructive*—they recover measured geometry and appearance, adding only learned priors where measurements are missing.

Table 4.1: Comparison of the reviewed methodologies

Aspect	**3D-RE-GEN**	**HouseCrafter**	**GS SLAM**	**Behind the Veil**
Paradigm	Generative, compositional	Generative, autoregressive	Reconstructive, online	Reconstructive with completion
Representation	Textured meshes	RGB-D views fused to TSDF mesh	3D Gaussians	Octree features and SDF
Learned prior	Pretrained detection, editing, and 3D models	Fine-tuned Stable Diffusion	None	Inpainter trained on complete rooms
Training needed	None	Large synthetic house set	None	Small set of complete rooms
Separate objects	Yes	No	No	No
Colour or texture	Yes	Yes	Yes	No
Speed	Minutes per scene	Many diffusion steps	Live	Per-scene optimisation

4.2 Input Requirements and Representations

The methods form a spectrum of input effort. 3D-RE-GEN needs only one photograph, and HouseCrafter only a floorplan, so both suit early design stages when no scan exists. Gaussian

Splatting SLAM needs a walkthrough video, and Behind the Veil needs posed depth images, so both suit capturing an existing home. Less input means more of the output is *imagined*: a single image cannot show the back of a wardrobe, and a floorplan says nothing about upholstery, so the generative methods trade fidelity to the real space for convenience [1, 2]. Representations follow from the goal: meshes for editing and export (3D-RE-GEN, HouseCrafter), Gaussians for fast, photorealistic viewing (Gaussian Splatting SLAM), and neural SDFs for clean, complete geometry (Behind the Veil) [3, 4].

4.3 Editability and Completeness

Virtual staging requires two properties: objects that can be moved independently, and surfaces that remain complete after they move. Only 3D-RE-GEN produces separate object meshes, and its 4-DoF placement keeps them on the floor [1]. HouseCrafter's fused house is one mesh, although the floorplan itself can be edited and the house regenerated [2]. Gaussian Splatting SLAM's map is easy to modify locally because Gaussians are independent primitives, but it has no object structure [3]. Behind the Veil is the only method that explicitly fills in surfaces hidden behind and under furniture, which is exactly what is revealed when an object is moved [4]. The works are therefore complementary: object decomposition from [1] combined with occluded-surface completion from [4] would yield fully editable rooms.

4.4 Real-Time Versus Offline Operation

Gaussian Splatting SLAM is the only live system, mapping at about 3 frames per second and rendering at hundreds, which suits AR previews and walkthroughs [3]. 3D-RE-GEN takes 7–20 minutes per room, HouseCrafter runs many diffusion batches, and Behind the Veil optimises per scene [1, 2, 4]; this suits one-off model creation, but interactive editing needs faster, feed-forward variants. Because each work uses different data and metrics, their numbers cannot be compared directly, and generative methods also rely on user studies [1, 2].

4.5 Strengths and Limitations

Table 4.2 summarises the main strengths and limitations identified in Chapter 3.

Table 4.2: Strengths and limitations of the reviewed works

Work	Strengths	Limitations
3D-RE-GEN	Separate textured objects and background; no training; objects grounded on the floor	Minutes per scene; relies on large proprietary models; hidden parts are imagined
HouseCrafter	Whole houses from a familiar floorplan; metric depth; explicit mesh	Trained on synthetic houses; single fused mesh; floorplan omits style and height
Gaussian Splatting SLAM	Live tracking and photorealistic mapping; compact; real-time rendering	No loop closure; no surface mesh or objects; unseen areas stay empty
Behind the Veil	Completes hidden surfaces; generalises from few rooms; clean meshes	Geometry only; needs posed depth; not real-time

4.6 Towards a Combined Home Design Workflow

Figure 4.6.1 sketches how the four methods could form one pipeline. A floorplan is lifted into an initial house model [2]; once the home exists, a walkthrough video produces a live photorealistic map [3], and its depth is completed behind and beneath furniture [4]; finally, individual pieces captured in photographs are turned into separate, ground-aligned assets that designers can move, replace, and restyle [1].

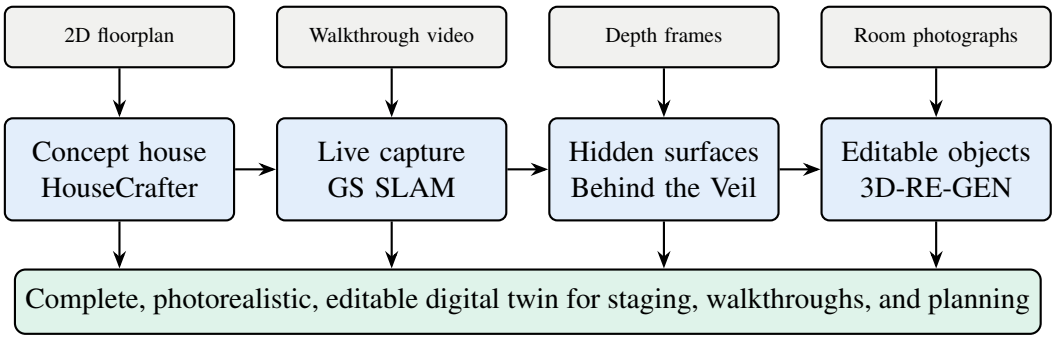


Figure 4.6.1: A combined home design workflow

Chapter 5

Applications

Neural 3D scene representations open many applications in the home and interior sector.

5.1 Virtual Staging and Interior Design

Designers can turn a client's photograph into an editable 3D room, remove or replace furniture, and render the result from new viewpoints [1]. Completed occluded surfaces mean that floors and walls stay intact when pieces are moved [4], and ground-aligned placement keeps new items physically plausible.

5.2 Real Estate Walkthroughs

Agents can capture a property with a phone and publish a photorealistic, real-time walkthrough built by Gaussian Splatting SLAM [3], while developers can present unbuilt apartments as complete 3D houses generated from their floorplans [2].

5.3 Architectural Digital Twins and Home Automation

Complete, accurate room models form digital twins for facility management, renovation planning, and smart homes, where devices and sensors are placed in a 3D model of the space. Complete geometry also lets home robots, such as cleaners, plan around and beneath furniture [4].

5.4 AR, VR, and E-Commerce

Furniture retailers can let customers place products in a 3D model of their own room in AR, and VR users can explore generated homes and interact with them, for example by placing a tablecloth or letting virtual agents navigate inside [2].

Chapter 6

Challenges and Future Directions

6.1 Open Challenges

- **Fidelity versus imagination:** generative methods invent hidden content that may not match the real home, which matters for measurements and purchases [1, 2].
- **Speed:** most methods take minutes or longer per room; only SLAM runs live, and only on a high-end GPU [3].
- **Semantics and editability:** few outputs separate objects, materials, and lighting, which design tools need [1].
- **Scale and drift:** whole homes need loop closure and consistent multi-room models [2, 3].
- **Data and evaluation:** complete real indoor scenes with ground truth are rare, and generative quality lacks reliable metrics [2, 4].

6.2 Future Directions

- **Unified pipelines** that combine live Gaussian capture with object decomposition and occluded-surface completion.
- **Feed-forward models** that reconstruct or generate a room in seconds rather than minutes.
- **Material and lighting decomposition**, so that staged rooms can be relit and refurnished realistically.
- **Language and layout control**, letting users restyle rooms by text while respecting floorplans.
- **On-device capture** on phones and AR headsets for consumer use.

Chapter 7

Conclusion

This seminar examined how Neural Radiance Fields and related neural 3D representations, especially 3D Gaussian Splatting, are transforming the creation of indoor 3D environments for home and interior design. After introducing scene representations, volume rendering, Gaussian splatting, signed distance fields, diffusion models, and visual SLAM, it reviewed four recent works in depth.

3D-RE-GEN reconstructs an editable room with separate, ground-aligned furniture and a complete background from a single image, using Application-Querying and a 4-DoF planar optimisation [1]. **HouseCrafter** lifts a 2D floorplan into a multi-room house by generating consistent RGB-D views with a floorplan-guided diffusion model and fusing them into a mesh [2]. **Gaussian Splatting SLAM** builds a photorealistic Gaussian map live from a single camera with analytic pose Jacobians and isotropic regularisation [3]. **Behind the Veil** completes the surfaces hidden behind and beneath furniture with a coarse-fine octree and a cross-scene 3D Inpainter [4].

Together, these works show that the bottlenecks of legacy CAD modelling and photogrammetry, namely manual effort, expensive capture, and incomplete, non-editable meshes, can be overcome by learned 3D priors. They are complementary: generative methods minimise input, reconstructive methods ensure fidelity, and their combination points towards complete, photorealistic, and editable digital twins of homes. Faster and unified pipelines that also model materials and lighting will make such tools practical for everyday virtual staging, real-estate walkthroughs, and automated spatial design.

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